

Computer Vision Assignment: Pallet Pose Estimation & Load Compliance

Role: AI/ML Engineer (Computer Vision) — Delhivery **Window:** 5 days. **Submit:** private GitHub repo + [README.md](#) + dataset + trained weights (or link) + a 5-min screen recording for the entire pipeline.

Problem

A warehouse staging area holds loaded pallets waiting for outbound dispatch. The floor is marked into numbered slots, and pallets are supposed to sit squarely within them — misaligned pallets block aisles, obstruct neighbouring slots, and cause disputes when a load is later found damaged and nobody can establish how it was staged.

A camera is mounted on a pillar overlooking the area, roughly 1.2m above the floor and tilted down about 20°.

Two things need to come out of each image:

Where each pallet is — position on the floor in metres and its orientation, so placement can be checked against the marked slot.

Whether the load complies with the warehouse's stacking standard, so a non-compliant pallet is caught while it is still in staging rather than after dispatch.

There are no fiducial markers. No ArUco, no AprilTags, no reflective targets. Nobody is going to maintain markers on 4,000 pallets in a working warehouse. Pose must come from the pallet's own geometry.

Slot compliance is assessed against a tolerance of **$\pm 2\text{cm}$ in position and $\pm 3^\circ$ in orientation**. That is the bar the pose has to clear to be useful.

No dataset is provided. Sourcing it is part of the work.



Just for reference

Section 1 — Dataset & Detection Model (30%)

Build a dataset and train a model that detects pallets and localises the geometry needed for pose.

Deliver:

- The dataset, in a standard format, plus your annotation tooling
- **DATASET.md** covering: where the data came from and why; what your sourcing choice cost you; class and annotation counts; your split protocol and what you split by; a one-page labelling guideline; known biases and gaps
- A trained model and its weights
- Detection and localisation accuracy reported **separately**, as distributions, on a held-out set that differs from your training data in a way you declare
- Your training decisions and the reasoning behind them
- Your estimate of the accuracy ceiling given your data volume, and what would raise it
- Sample Datasets :-

<https://app.roboflow.com/rj-xvfw4/plh-c-1-veibf/1>

<https://app.roboflow.com/rj-xvfw4/pallet-ff5lh-fp7tw/1>

Section 2 — Pose Estimation (35%)

Produce metric pose for each detected pallet: **position (x, y) in metres in a floor frame you define, orientation θ , and which face is which** — pallets are not rotationally symmetric in use, and the report needs to distinguish a pallet at 30° from one at 45°.

Deliver:

- Camera calibration artefacts and reprojection error
 - Your pose method and the reasoning behind it
 - **An evaluation you construct yourself** — there is no pose ground truth unless you make it. Justify how you built it.
 - Error reported as distributions, **translation and rotation separately**
 - Sensitivity of your pose to error in your camera height and tilt estimates, at short and long range
 - The **usable envelope**: at what range and viewing angle does your pose actually meet the $\pm 2\text{cm}$ / $\pm 3^\circ$ bar, and where does it not
 - What your system outputs when it cannot produce a reliable pose
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Section 3 — Load Analysis & SOP Checks (25%)

Assess whether the load on the pallet complies with the standard below.

None

SOP-PAL-03: Pallet Load Standard

1. No box overhangs the pallet edge by more than 3cm.
2. Load height not to exceed 1.8m from floor.
3. Boxes stacked in aligned columns; no box rotated more than 15° relative to the pallet's principal axes.
4. Larger boxes below smaller boxes; no size inversion.
5. Load stretch-wrapped before dispatch.
6. No visibly damaged or crushed box in the load.
7. Load centred on pallet; centroid within 10cm of pallet centre.
8. Pallet itself undamaged – no broken boards or split stringers.

The camera sees each pallet from one side only.

Deliver:

- A triage of all eight steps: which are verifiable from this view, which are partially verifiable and under what assumptions, which are not verifiable at all — with justification for each
 - Implementation of the subset you judge verifiable, and your reasoning for that subset
 - A per-check confidence, and an explanation of what it is derived from
 - A **pass / fail / manual-inspection** verdict per pallet with justified thresholds, and your reasoning for how you weighted pose quality against load compliance
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Section 4 — Deployment & Robustness (10%)

Mostly written. Target hardware is a Jetson Orin Nano at 15W; target throughput is ≥ 15 FPS.

Deliver:

- A latency analysis, with the hardware you actually measured on clearly stated
 - What you would expect to change on the target hardware, and why
 - The accuracy cost of quantisation, if you export — measured on whatever matters most for this task
 - How the system behaves when it fails, and how a downstream consumer knows it has failed
 - How you would use the fact that a pallet sits in the same place across many frames
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Required Output Format

One assessment per pallet, containing at minimum: pose with uncertainty, orientation, per-step SOP verdicts with confidence, and the overall verdict with its reasoning.

Design the format yourself. It should be something an engineer investigating a disputed assessment could actually use.

Rubric

Criterion	Weight
Dataset & detection	30%
Pose estimation	35%

Criterion	Weight
Load analysis & SOP	25%
Deployment reasoning	10%

Within each, we weight: correctness of the geometric reasoning, honesty of the evaluation, quality of judgement about what is and isn't possible, and whether someone else could read your report and understand what you built.

README — required sections

1. Approach and the significant decisions, with what each cost you
 2. Results — distributions, not point estimates
 3. Failure analysis — your three worst cases, with images, root-caused
 4. What you couldn't finish and why — **explicitly rewarded**
 5. AI tool usage — which tools, for what, and one thing they got wrong that you caught
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Ground rules

- Any open-source model, library, or dataset. Cite everything.
- AI assistants expected and allowed — disclose per README section 5.
- **The spec is underspecified on purpose.** Decide, document, move on. Don't email us to ask.
- **Confident wrong answers are the worst possible outcome.** These verdicts have consequences for people. Where you are uncertain, say so and quantify it.
- Reporting benchmark numbers from hardware you do not own is an automatic fail. "I don't have one, here's my reasoning" is completely fine.
- **Don't exceed the time budget.** Scoping is being evaluated. A finished 70% submission beats an abandoned 95% one.

Follow-up (45 min live)

We'll open your code and change a requirement. We're testing whether your design survives it, and whether you understand what you submitted.

Good Luck